

RESEARCH & ANALYTICS PORTFOLIO

ECONOMIC UNCERTAINTY IN INDIA

Measuring Its Time-Varying Impact on the Economy & Financial Markets

ADITI KUSHWAHA

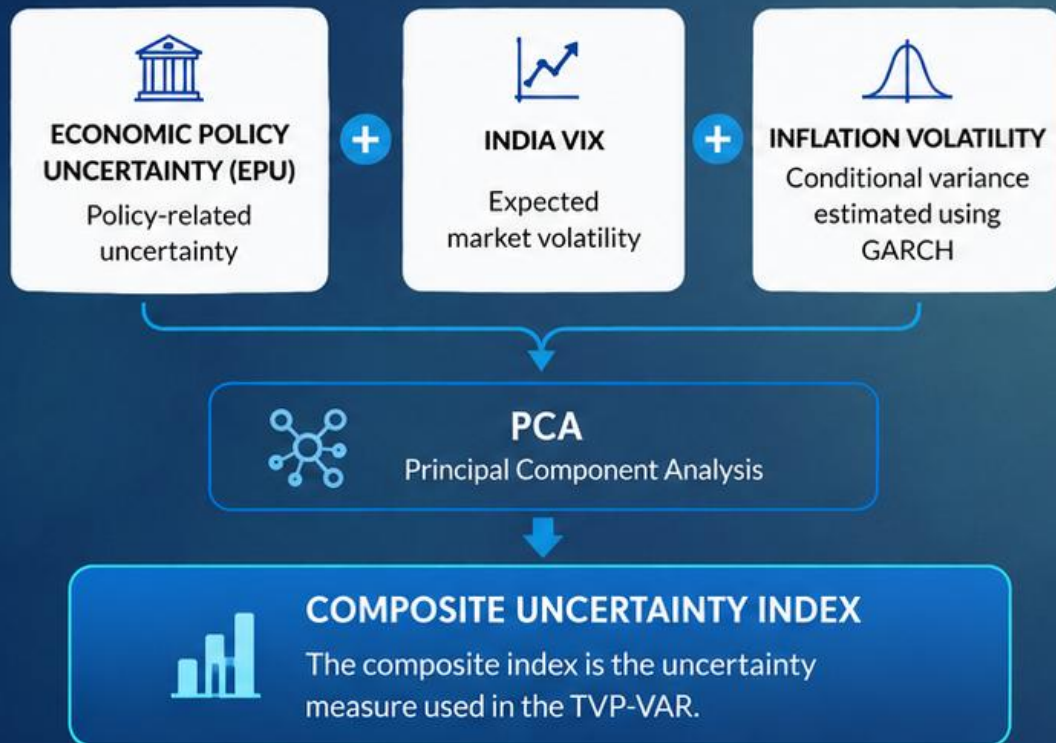
PROJECT AT A GLANCE



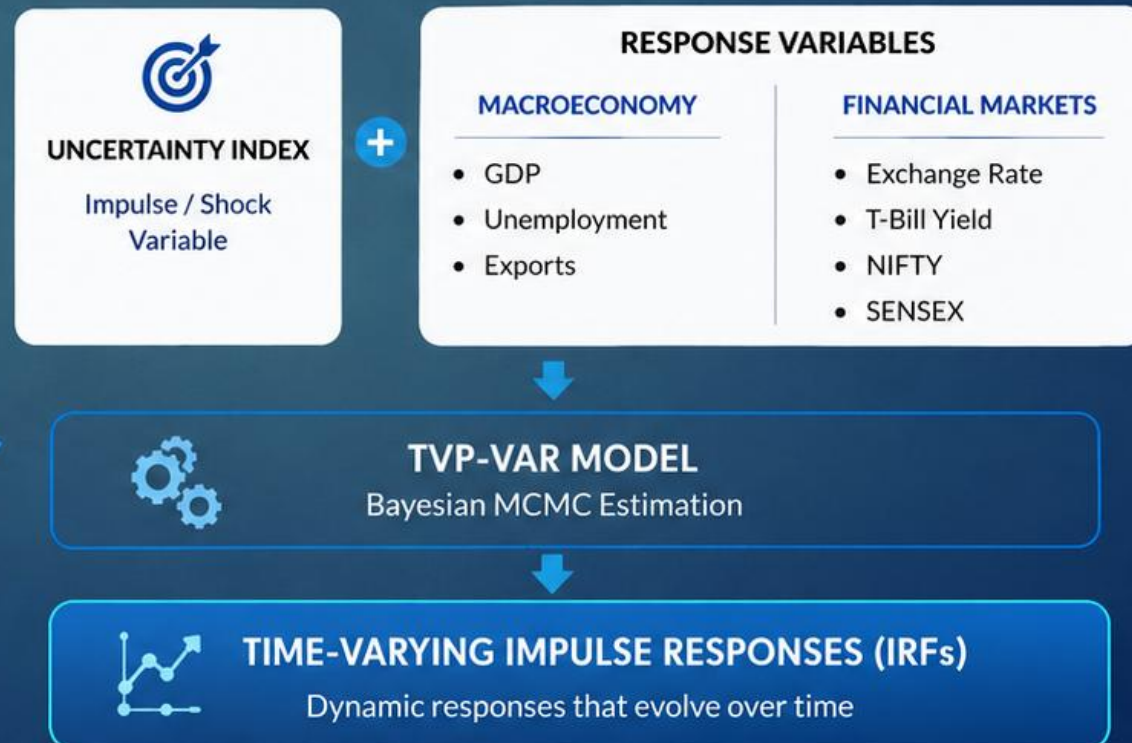
WHAT IS BEING ANALYSED?

How does an uncertainty shock affect India's economy and financial markets and how does the impact change over time?

01 CONSTRUCTING THE UNCERTAINTY INDEX



02 TVP-VAR ANALYSIS



2016-2022



MONTHLY DATA

PRE-COVID

COVID

POST-COVID

DATA & VARIABLES

2016–2022 | Monthly Indian Data

- Monthly data, 2016–2022 (includes pre-pandemic, pandemic and post-pandemic periods).
- GDP is a quarterly series converted to monthly frequency using the **Denton–Cholette** method with the **Index of Industrial Production (IIP)** as benchmark.

Stationarity treatment:

First differences applied to

Exchange Rate, Exports, GDP, Unemployment Rate and T-Bill Yield.

01 UNCERTAINTY MEASURES

ECONOMIC POLICY UNCERTAINTY (EPU)

Policy-related uncertainty measured from newspaper coverage.

INDIA VIX

Expected market volatility over the next 30 days based on NIFTY 50 options.

INFLATION VOLATILITY

CPI inflation series; conditional variance estimated using GARCH model.

Composite Uncertainty Index

EPU + India VIX + Inflation Volatility
→ PCA → **PC1 (Composite Index)**

02 VARIABLES ANALYSED

MACROECONOMIC VARIABLES

- GDP (Gross Domestic Product)
- Unemployment Rate
- Exports (Merchandise Exports)

FINANCIAL MARKET VARIABLES

- Exchange Rate (INR / USD)
- T-Bill Yield
- NIFTY (%)
- SENSEX (%)

All series are monthly.
Percent returns for NIFTY and SENSEX;
percentage for Unemployment, T-Bill Yield
and Inflation.

03 DATA SOURCES & UNITS

Variable	Source	Unit
Unemployment Rate	CMIE	Percentage
Exchange Rate	Financial Benchmarks India Pvt. Ltd. (FBIL)	INR / USD
Exports	Ministry of Commerce	Rs. Lacs
GDP	Ministry of Statistics & Programme Implementation (MoSPI)	Rs. Crore
T-Bill Yield	Financial Benchmarks India Pvt. Ltd. (FBIL)	Percentage
NIFTY (%)	Yahoo Finance	Percentage
Inflation	NSO, MoSPI	Percentage
SENSEX (%)	BSE India	Percentage
EPU	www.PolicyUncertainty.com	Index
India VIX	NSE India	Index

METHODOLOGY OVERVIEW

Goal: Construct a Composite Uncertainty Index and analyse its macro-financial impacts
2016–2022 | Monthly Indian Data

01

DATA PREPARATION

- Collect monthly data for all variables.
- Handle missing values, align frequencies and transform where required.

Frequency: Monthly, 2016–2022.

GDP (quarterly) converted to monthly using **Denton–Cholette** method with the **Index of Industrial Production (IIP)** as benchmark.

02

UNCERTAINTY MEASURES

- Compute Economic Policy Uncertainty (EPU), India VIX and Inflation Volatility (GARCH conditional variance).

03

COMPOSITE UNCERTAINTY INDEX (PCA)

- Standardise the three uncertainty measures.
- Apply Principal Component Analysis (PCA).
- Use the first principal component (PC1) as the Composite Uncertainty Index.

Inputs: EPU • India VIX • Inflation Volatility | **Method:** Standardization → PCA
Output: PC1 (Composite Uncertainty Index)

04

TVP-VAR MODEL

- Estimate a Time-Varying Parameter VAR (TVP-VAR) with the Composite Uncertainty Index and macro-financial variables.

05

ANALYSIS & INTERPRETATION

- Analyse time-varying impulse responses and forecast error variance decompositions to assess macro-financial impacts over time.

WHAT THIS ACHIEVES

- Captures multiple dimensions of uncertainty in a single index.
- Allows uncertainty effects to evolve over time.
- Provides insights into macroeconomic and financial channels.

KEY INPUTS

- **Uncertainty Measures:** EPU, India VIX, Inflation Volatility
- **Macroeconomic Variables:** GDP, Unemployment, Exports
- **Financial Market Variables:** Exchange Rate, T-Bill Yield, NIFTY, SENSEX

MODEL SPECIFICATION (TVP-VAR)

- Time-varying coefficients modeled as random walks
- Bayesian estimation using Markov-Chain Monte Carlo (MCMC)
- 60,000 iterations with 30,000 burn-in

IDENTIFICATION

Impulse (shock):
Uncertainty Index
(One-standard-deviation increase)

RESPONSES

GDP, Unemployment, Exports,
Exchange Rate, T-Bill Yield,
NIFTY and SENSEX

EXPECTED OUTPUT

- Composite Uncertainty Index (monthly)
- Time-varying impulse responses (IRFs) for 6- and 12-month horizons
- Variance decompositions (FEVD)
- Insights for policy and market participants

Note: IRFs are estimated using the TVP-VAR methodology and reflect responses over time.
The charts show point estimates. Confidence intervals are not displayed in the portfolio charts.



PRE-COVID

(Jan 2016 – Feb 2020)



COVID

(Mar 2020 – Feb 2022)



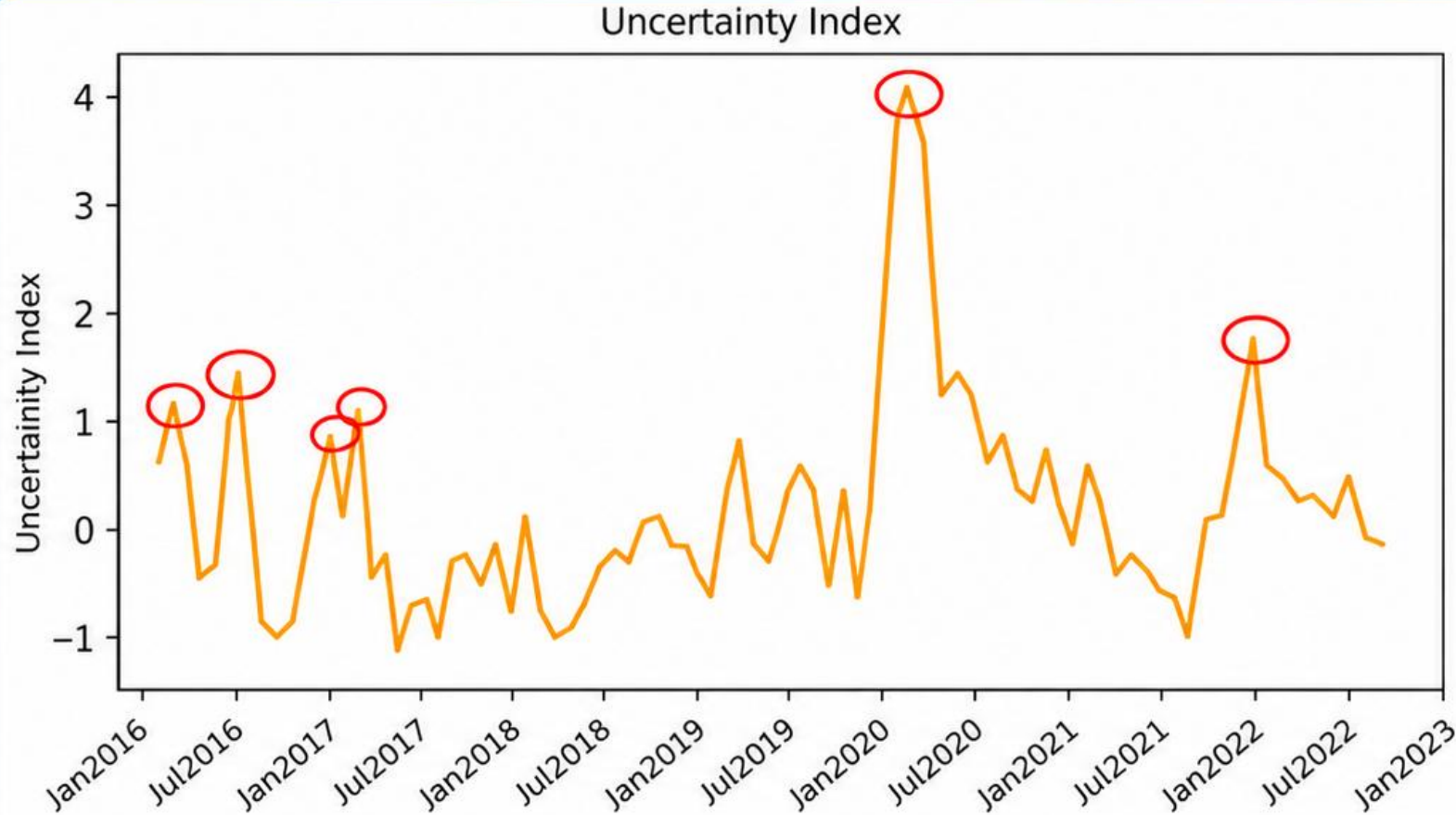
POST-COVID

(Mar 2022 – Dec 2022)

ANALYSIS AND RESULTS

Uncertainty Index

The figure shows the evolution of the composite Uncertainty Index in India, highlighting major spikes over time.



KEY OBSERVATIONS

- 1 **2020 marks an exceptional uncertainty shock**, with the index reaching its highest level in the sample.
- 2 **Uncertainty is strongly episodic**, with smaller spikes in 2016–17 and 2022, separated by relatively calmer periods (notably 2018–19).
- 3 **Post-2020 volatility persists**, although uncertainty gradually moves back toward lower levels.



CONCLUSION

- ✓ Uncertainty in India is highly time-varying: major shocks create sharp surges, followed by partial normalization rather than a complete return to pre-shock conditions.



2016–2022



MONTHLY DATA

PRE-COVID

COVID

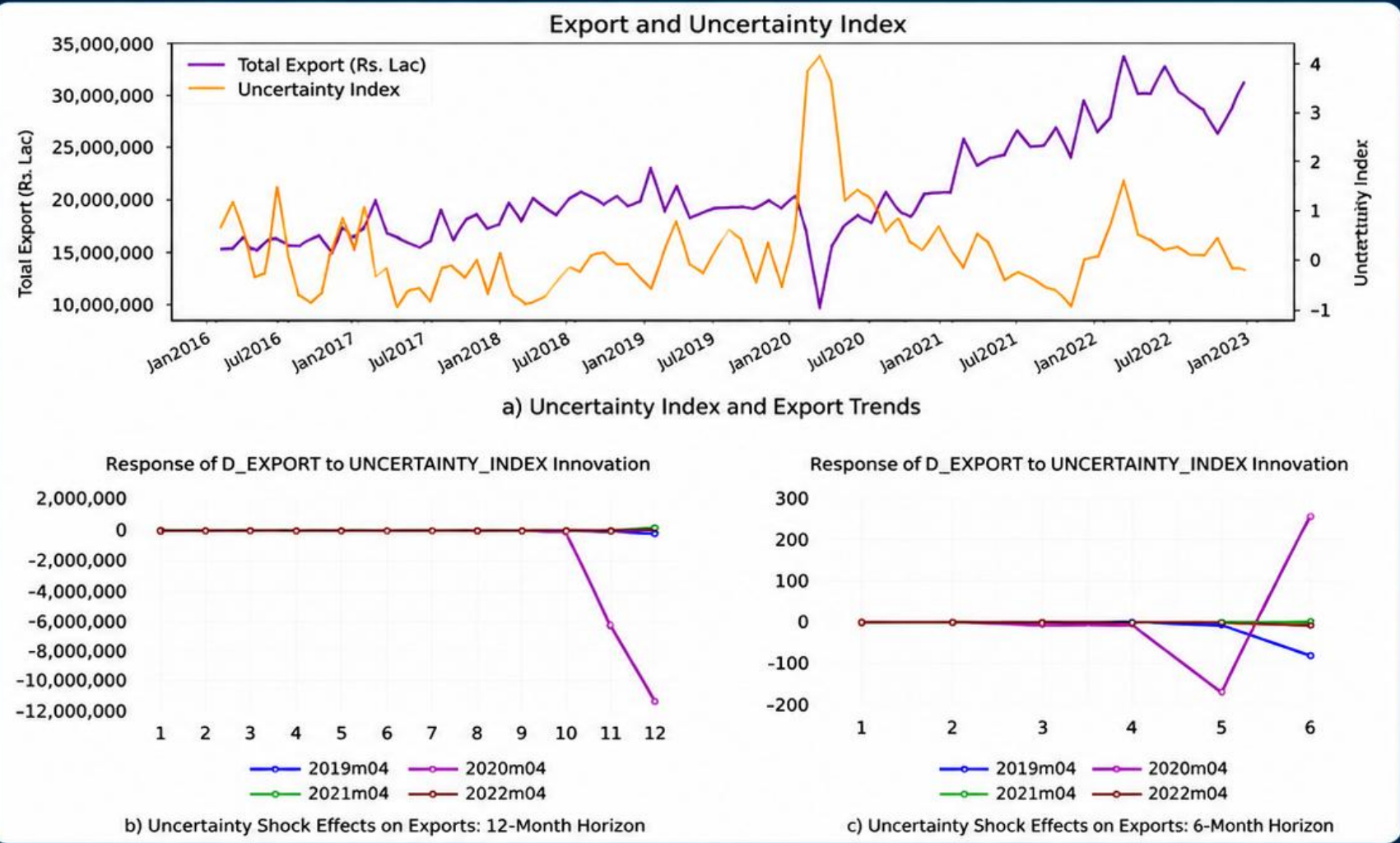
POST-COVID

Data for a More Resilient Economy

ANALYSIS AND RESULTS

Export and Uncertainty Index

The figure shows the co-movement between India's total exports and the Uncertainty Index, and the impulse response of exports to uncertainty shocks.



KEY OBSERVATIONS

- Exports show an overall upward trend, with a sharp dip in 2020 coinciding with a spike in uncertainty.
- Uncertainty and exports generally move inversely, especially during crisis periods.
- 12-month horizon: a large negative and persistent response, particularly for 2020m04.
- 6-month horizon: an initial negative response (notably 2020m04), followed by a positive rebound by month 6.

CONCLUSION

- The estimated export response moves below baseline following the uncertainty shock, with the largest response concentrated around the 2020 episode. The response varies across horizons.



2016-2022



MONTHLY DATA

PRE-COVID

COVID

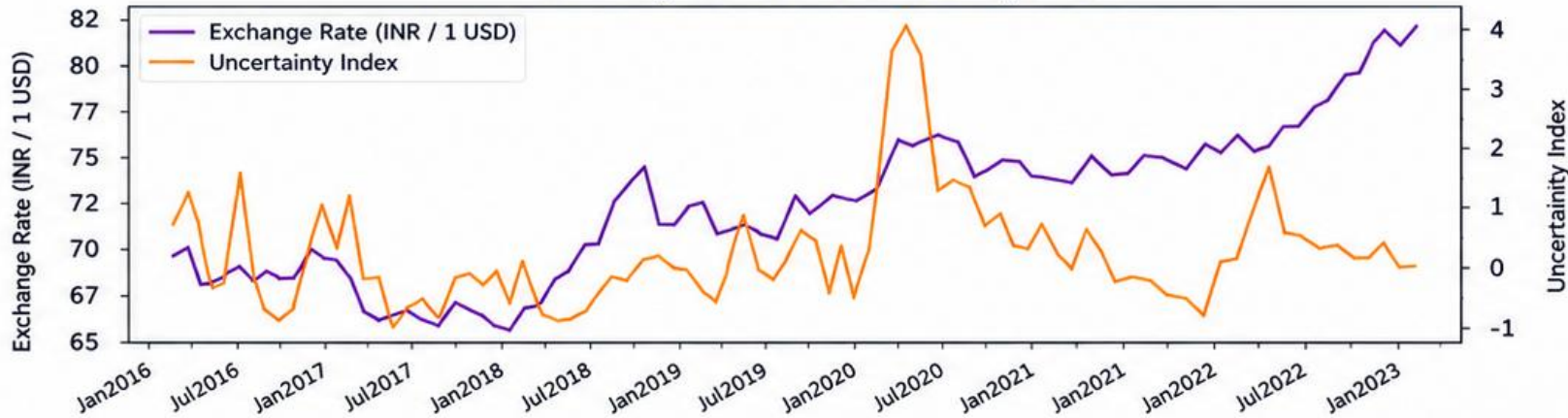
POST-COVID

ANALYSIS AND RESULTS

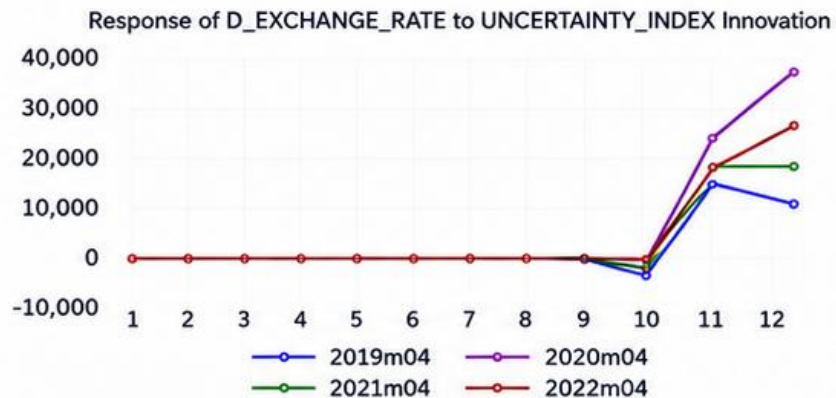
Exchange Rate and Uncertainty Index

The figure shows the co-movement between India's exchange rate and economic uncertainty, and the impulse response of the exchange rate to uncertainty shocks.

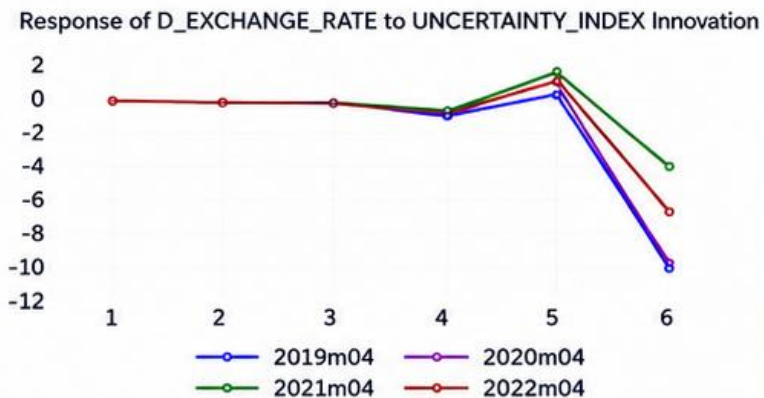
Exchange Rate and Uncertainty Index



a) Uncertainty Index and Exchange Rate Trends



b) Uncertainty Shock Effects on Exchange Rate: 12-Month Horizon



c) Uncertainty Shock Effects on Exchange Rate: 6-Month Horizon

KEY OBSERVATIONS

- Exchange rate and uncertainty **move together** overall, with a pronounced uncertainty spike in early 2020.
- The exchange rate shows **depreciation** over time, with faster depreciation during periods of higher uncertainty.
- Uncertainty moderates after 2020, while **depreciation pressures continue and intensify in 2022**.

IMPULSE RESPONSE RESULTS

- 12-month horizon: the estimated response indicates a **pronounced increase** in the exchange rate (depreciation), especially around the 2020m04 and 2022m04 episodes.
- 6-month horizon: the initial response is small, followed by a **positive response that peaks around month 5** and then adjusts in month 6.

CONCLUSION

- The estimated responses suggest that uncertainty shocks are associated with higher exchange rate (depreciation) effects that are **larger and more persistent** during major crisis episodes.



2016-2022



MONTHLY DATA

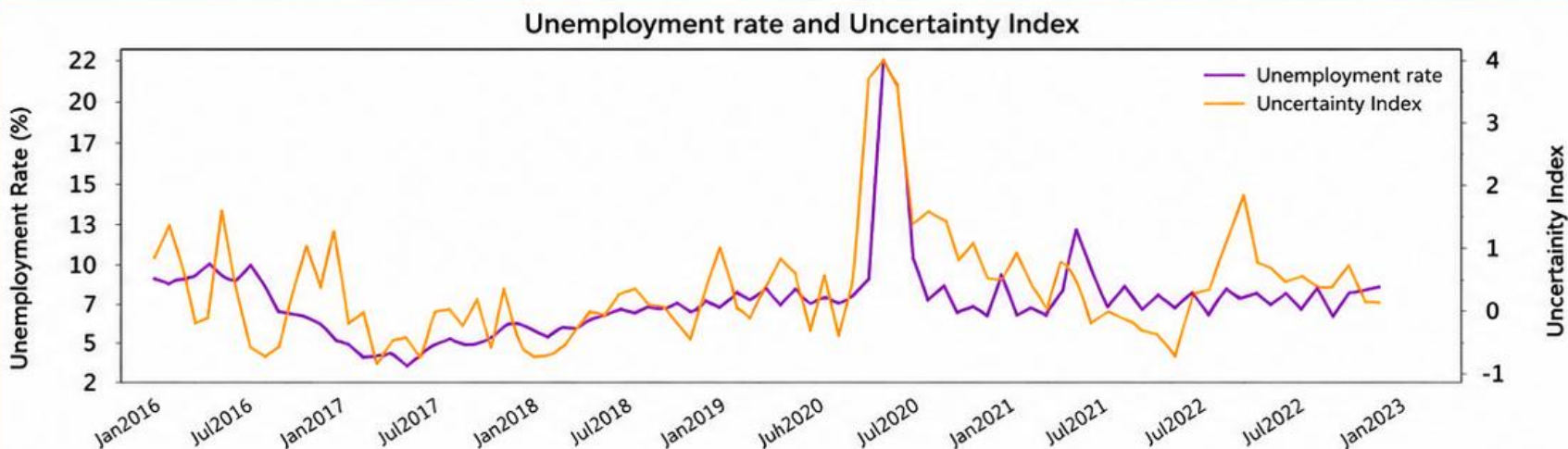
PRE-COVID

COVID

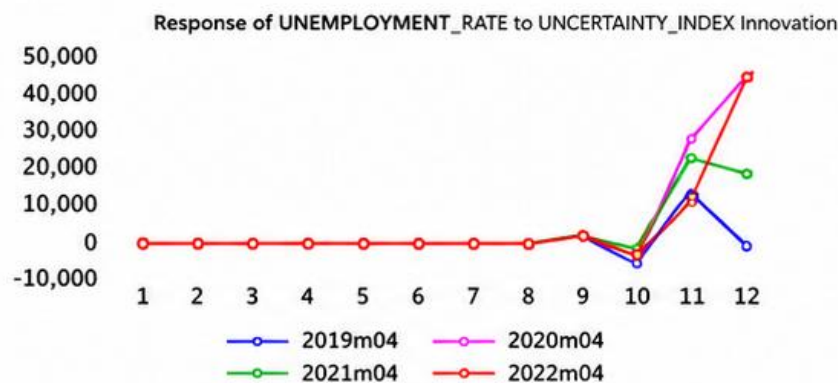
ANALYSIS AND RESULTS

Unemployment Rate and Uncertainty Index

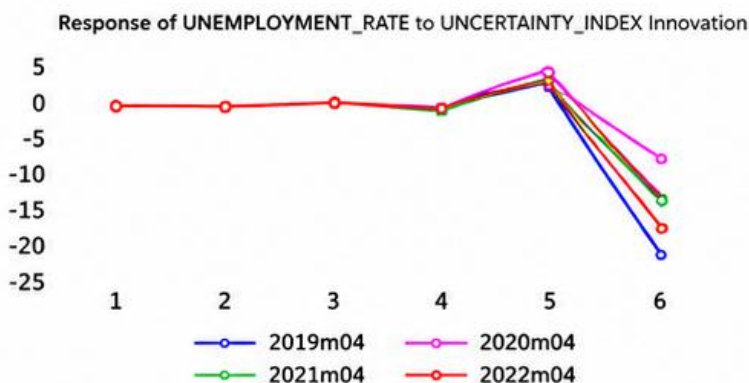
The figure shows the co-movement between India's unemployment rate and the Uncertainty Index, and the impulse response of unemployment to uncertainty shocks.



a) Uncertainty Index and Unemployment Rate Trends



b) Uncertainty Shock Effects on Unemployment Rate: 12-Month Horizon



c) Uncertainty Shock Effects on Unemployment Rate: 6-Month Horizon

KEY OBSERVATIONS

- Unemployment and uncertainty move together, with a sharp spike in 2020.
- Over 12 months, uncertainty shocks lead to a large and persistent rise in unemployment, strongest for 2020m04 and 2022m04.
- Over 6 months, the response is small initially and turns negative by month 6.
- Responses differ across episodes.

CONCLUSION

- A one-standard-deviation increase in the Uncertainty Index is associated with a higher unemployment response over the longer horizon, with greater persistence during major disruptions.

IMPLICATION

- Higher uncertainty can extend labour-market adjustment, underscoring the need for timely measures to support employment and economic activity.



2016-2022



MONTHLY DATA

PRE-COVID

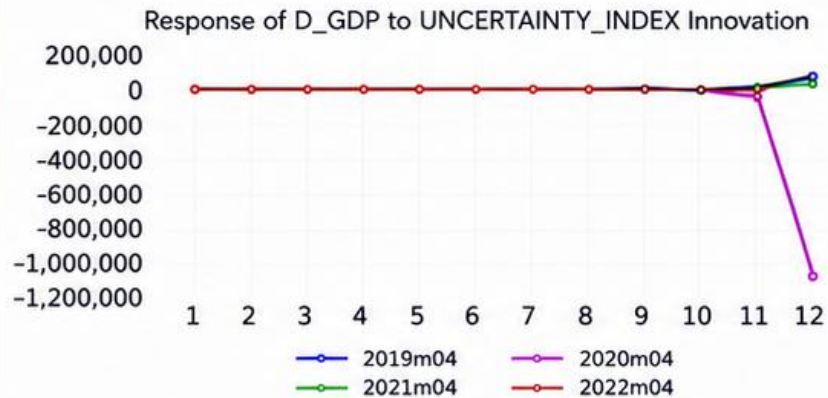
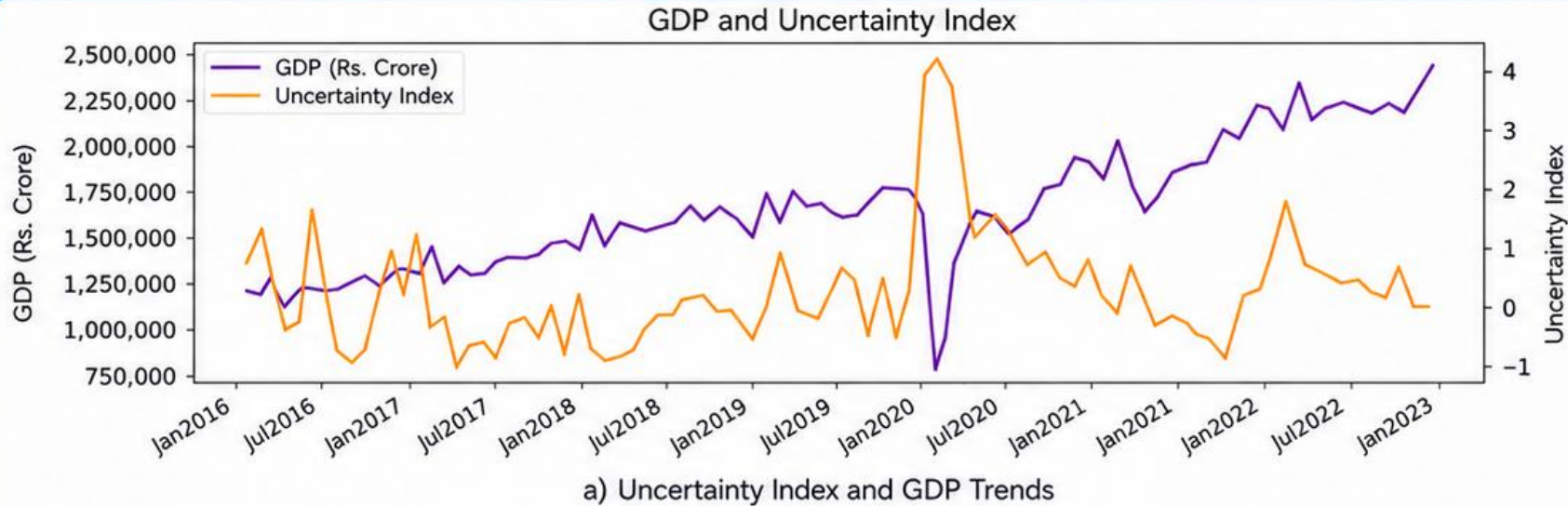
COVID

POST-COVID

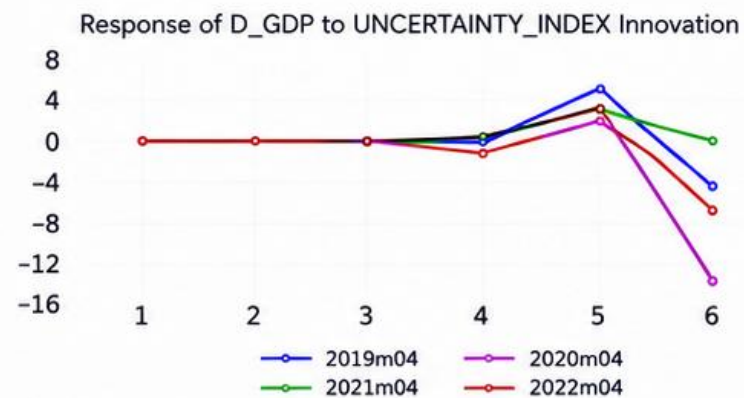
ANALYSIS AND RESULTS

GDP and Uncertainty Index

The figure shows the co-movement between India's GDP and the Uncertainty Index, and the impulse response of GDP to uncertainty shocks.



b) Uncertainty Shock Effects on GDP: 12-Month Horizon



c) Uncertainty Shock Effects on GDP: 6-Month Horizon

KEY OBSERVATIONS

- GDP shows a strong upward trend overall, with a sharp dip in 2020 during the pandemic.
- Higher uncertainty is followed by a short-term contraction in GDP.
- 12-month horizon: large and persistent negative impact on GDP, especially for 2020m04.
- 6-month horizon: initial positive response (months 4–5), followed by a sharp decline by month 6.

CONCLUSION

- Uncertainty shocks are associated with a notable decline in GDP over medium horizons, with the largest adverse effects observed during the COVID-19 period.



2016–2022



MONTHLY DATA



PRE-COVID



COVID

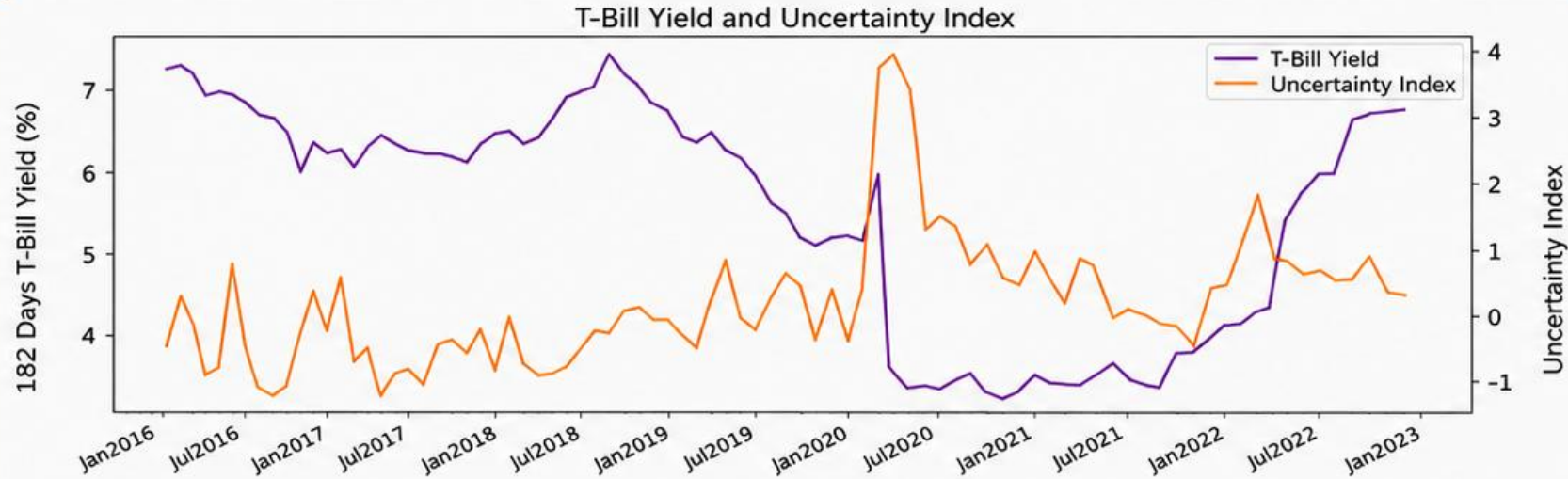


POST-COVID

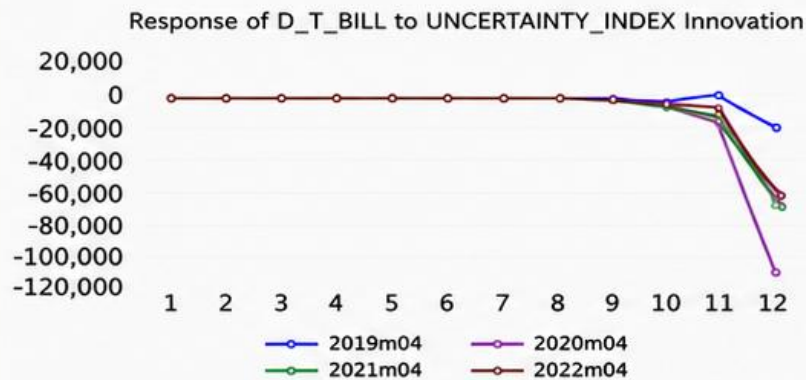
ANALYSIS AND RESULTS

T-Bill Yield and Uncertainty Index

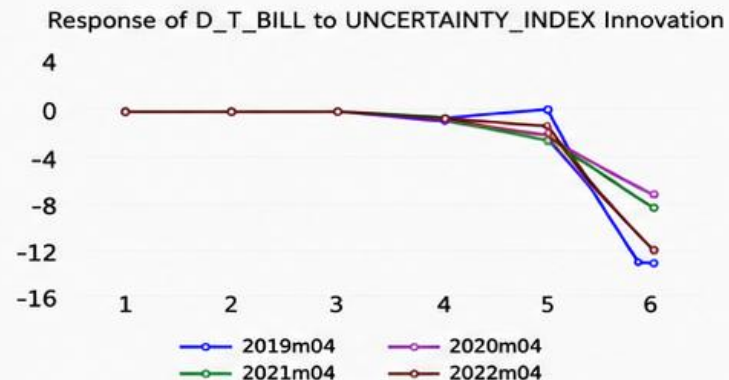
The figure shows the co-movement between India's 182-day T-Bill yield and the Uncertainty Index, and the impulse response of T-Bill yield to uncertainty shocks.



a) Uncertainty Index and T-Bill Yield Trends



b) Uncertainty Shock Effects on T-Bill Yield: 12-Month Horizon



c) Uncertainty Shock Effects on T-Bill Yield: 6-Month Horizon

KEY OBSERVATIONS

- T-Bill yields and uncertainty show a broadly inverse movement over the sample period.
- The 2020 uncertainty spike coincides with a pronounced decline in T-Bill yields.
- 12-month horizon: the estimated response remains negative, with the largest decline in 2020m04.
- 6-month horizon: the estimated response is negative, with notable declines in 2019m04 and 2022m04.

CONCLUSION

- A one-standard-deviation increase in uncertainty is associated with a lower T-Bill yield response, with larger and more persistent responses during major disruption episodes.



2016-2022



MONTHLY DATA



PRE-COVID



COVID

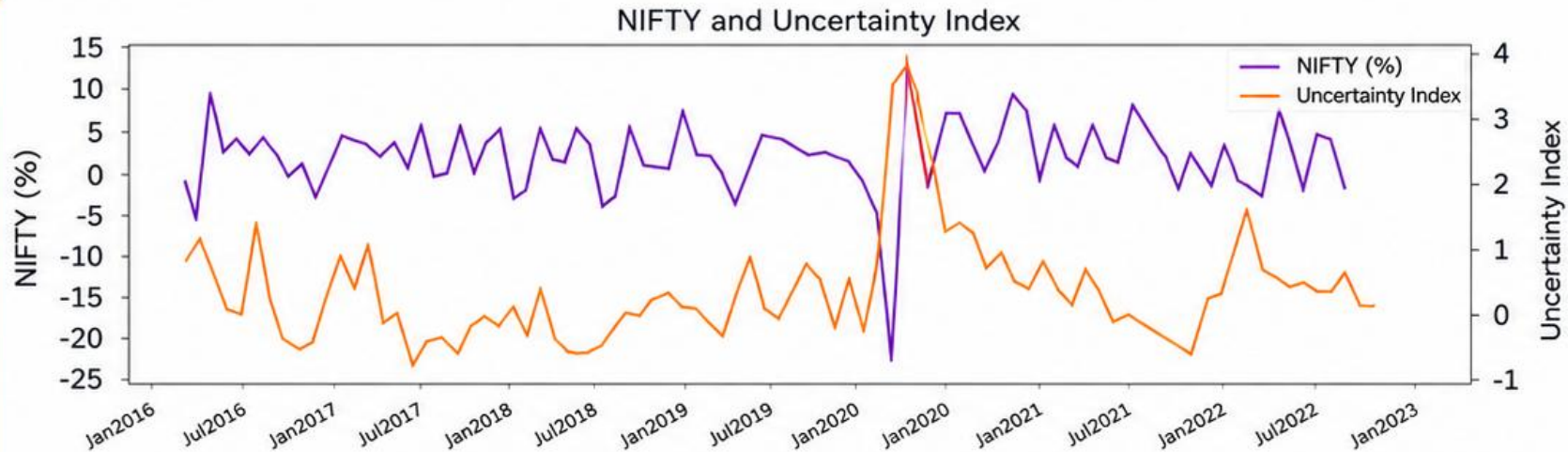


POST-COVID

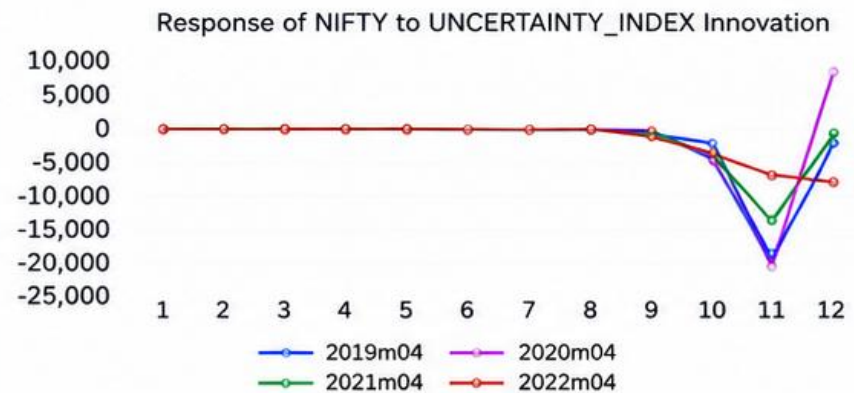
ANALYSIS AND RESULTS

NIFTY and Uncertainty Index

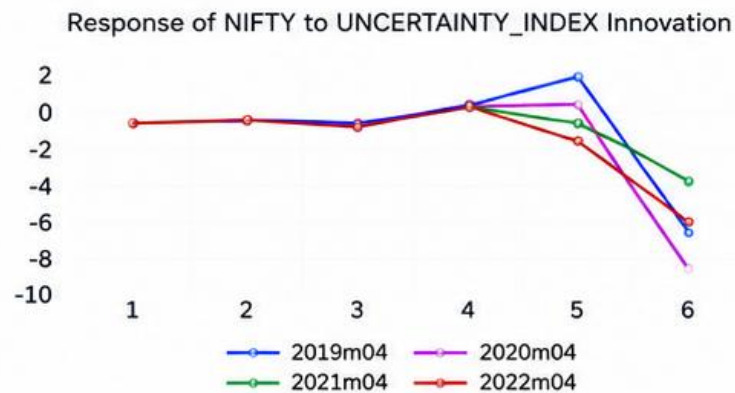
The figure shows the co-movement between NIFTY returns and the Uncertainty Index, along with the estimated impulse response of NIFTY to uncertainty shocks.



a) Uncertainty Index and Nifty Trends



b) Uncertainty Shock Effects on Nifty: 12-Month Horizon



c) Uncertainty Shock Effects on Nifty: 6-Month Horizon

KEY OBSERVATIONS

- NIFTY returns and the Uncertainty Index exhibit an inverse co-movement throughout the sample period.
- The sharp increase in uncertainty in 2020 coincides with a pronounced decline in NIFTY returns.
- Impulse response results indicate a negative response of NIFTY to uncertainty shocks across all horizons.
- The largest adverse impact is observed around month 4 in the 12-month horizon and around month 6 in the 6-month horizon.
- NIFTY shows signs of recovery after the initial decline, suggesting market resilience over longer horizons.

CONCLUSION

- Uncertainty shocks are associated with a short-term decline in NIFTY returns; however, the market tends to recover over time.



2016-2022



MONTHLY DATA



PRE-COVID



COVID

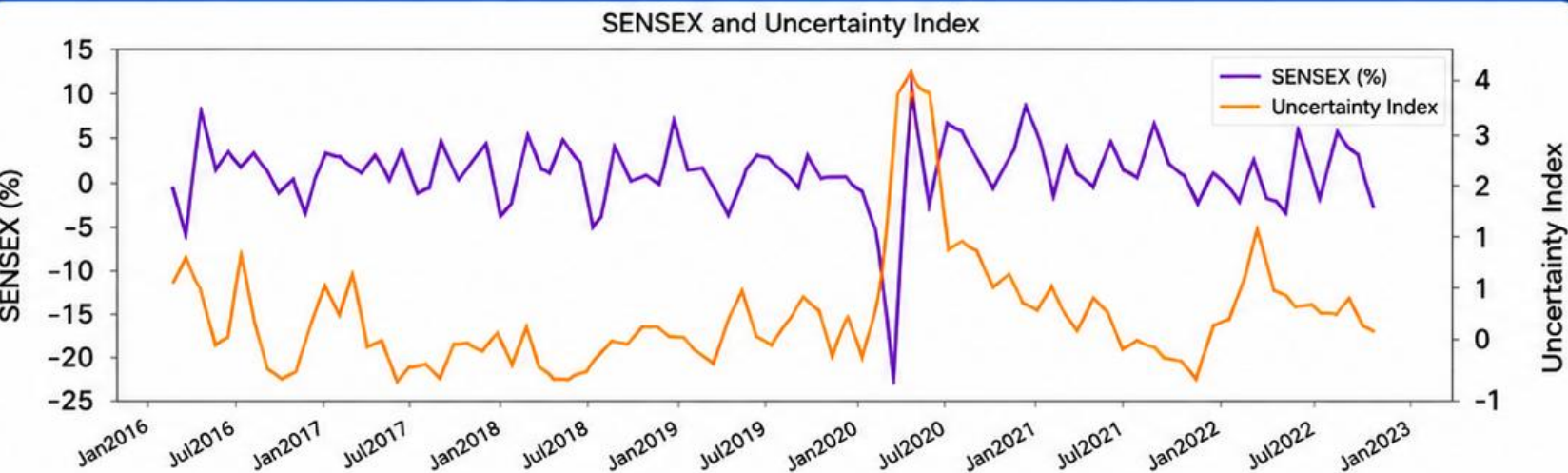


POST-COVID

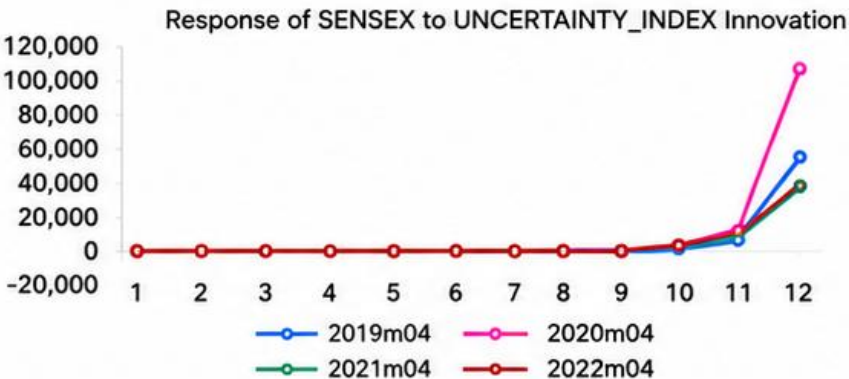
ANALYSIS AND RESULTS

SENSEX and Uncertainty Index

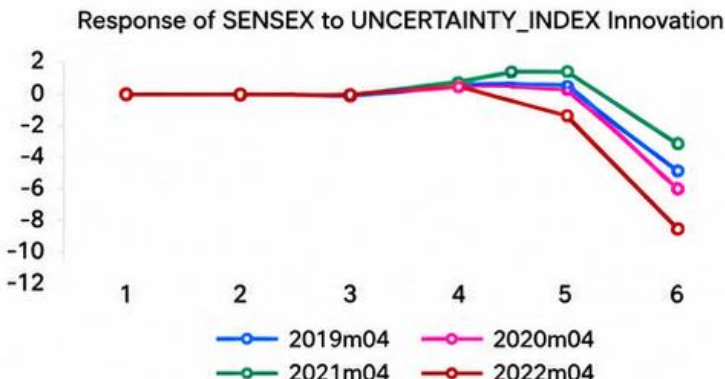
The figure shows the co-movement between SENSEX returns and the Uncertainty Index, and the impulse response of SENSEX to uncertainty shocks.



a) Uncertainty Index and SENSEX Trends



b) Uncertainty Shock Effects on SENSEX: 12-Month Horizon



c) Uncertainty Shock Effects on SENSEX: 6-Month Horizon

KEY OBSERVATIONS

- SENSEX returns and the Uncertainty Index move inversely, with a sharp market decline during the 2020 spike in uncertainty.
- Short-run (6-month) response is negative, with effects intensifying by month 6.
- Initial adverse impact in the 12-month horizon is followed by a recovery, with strong positive response for 2020m04.
- Markets show resilience and tend to recover over longer horizons.

CONCLUSION

- Uncertainty shocks cause short-term declines in SENSEX, but markets tend to rebound and show resilience over longer horizons.

IMPLICATION

- Financial markets are sensitive to uncertainty in the short run, but have recovery capacity over time.



2016-2022



MONTHLY DATA

PRE-COVID

COVID

POST-COVID

OVERALL RESULTS

Synthesis of the impact of economic uncertainty on India's economy and financial markets (2016–2022)

1 ECONOMIC ACTIVITY

Uncertainty weakened key macroeconomic outcomes.

- **GDP:** the estimated response moves below baseline, with the largest decline during 2020.
- **Exports:** the estimated response moves below baseline, with the strongest and most persistent response around 2020.
- **Unemployment:** the estimated response moves above baseline, particularly during 2020.
- **T-Bill yield:** the estimated response moves above baseline, consistent with an inverse relationship.

2 FINANCIAL MARKETS

Uncertainty increased market stress and weakened returns.

- **NIFTY:** the estimated response moves below baseline, with the strongest short-run response around 2020.
- **SENSEX:** the estimated response moves below baseline, particularly during high-uncertainty periods.
- Market performance subsequently shows signs of adjustment and recovery across variables.

3 TIME-VARYING EFFECT

The impact was not constant across periods.

- **COVID-19 period:** the estimated responses are the largest and most widespread.
- **Post-COVID period:** several responses become weaker or move closer to baseline.
- **2022:** renewed uncertainty produces different response patterns across variables.

Overall, a **one-standard-deviation increase** in the Composite Uncertainty Index is associated with **dynamic responses** across India's economy and financial markets. The estimated responses are **strongest during the COVID-19 period** and vary across variables and time.

CONCLUSION & POLICY CONSIDERATIONS

CONCLUSION

- Economic policy uncertainty in India is **persistent and time-varying**, with sharp increases during major economic and policy disruptions, especially in the COVID-19 period.
- Higher uncertainty is associated with **weaker short-run economic activity** and **financial-market performance**, affecting GDP, industrial activity, exports, stock-market returns and employment.
- Uncertainty also affects **financial conditions**, through **exchange-rate depreciation** and changes in T-bill yields, indicating transmission via both currency and financial markets.
- The **effects are not uniform** across time and variables. The magnitude, direction and persistence of uncertainty shocks vary across sub-periods and horizons, with the strongest and most widespread effects during major crisis episodes.

POLICY CONSIDERATIONS

- **Enhance policy clarity and communication:** Provide clear, timely and consistent policy communication and forward guidance to reduce uncertainty and support predictable decision-making.
- **Safeguard external stability:** Maintain adequate foreign-exchange reserves and a stable trade policy to manage exchange-rate pressures and support exports during periods of high uncertainty.
- **Support activity and employment:** Deploy timely fiscal measures and targeted public investment to cushion the contraction in industrial activity and employment linked to uncertainty shocks.
- **Strengthen financial-market resilience:** Ensure credible monetary policy communication and adequate market liquidity to limit volatility in T-bill yields and equity markets during uncertain periods.



Economic uncertainty is a **key transmission channel** through which major shocks affect India's **macroeconomic performance** and **financial stability**, with the magnitude and persistence of its effects varying across time and variables.

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THANK YOU